Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

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Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

Structure of a BSc. / MSc. Project or Thesis in Computing Science / Information Science Work-in-progress Description

Your Name Version of December 16, 2019

1 Title of the Project / Thesis

- Descriptive title of the project / thesis; not too specific

2 Motivation and Research Question (Short Version of what is described below)

- Starting point / initial position
- Scientific and practical question
- Expected usage
- Result qualification / quantification / measuring
- Metrics for evaluating project success
- ...

3 Starting Position

- Preliminary work (own and related work)
- Assumptions
- Data (Source, Description, Type)
- ...

4 Originality and innovative character

4.1 Impact: Why is this research important for the Society and or Research Community

- State the project's overall objective and broad vision.
- "Added Value": What is the benefit of your project to the research field? Which parts open new new research questions or future work?
- From a conceptual point of view: what is new and important besides your implementation / system / concrete application?

4.2 Delineation: What will not be considered? And Why?

- Enumeration of special cases, algorithms, etc., which are not explicitly considered
- Delineation, show the "Novelty" part of the thesis in comparison to existing techniques

5 Goals

- Primary (Must-Have) Goals
  - Operationalized description of the main goals of this project and thesis (methodical and practical)
  - Deploy assessment criteria of your method: technical or methodical novelty, efficiency, effectiveness, etc.
  - Rough estimation of expenditure of time; list of possible risks (what is difficult? what might be irresolvable?) Which sub-projects may fail? Are there alternatives?
- Desired goals / Future work for bachelor / master thesis
  - What else can be considered in the project (with low priority)?
  - In which sub-projects can we (unexpectedly?) receive nice results?

6 Literature Research / Related Work

- Description of keywords used for the literature search
- Listing of known literature: books, survey papers, fundamental works, and current publications (state of the art approaches)
- Develop a mental-model and structure of the field

7 Working Schedule

- Write it down
- Milestones and time schedule

8 Literature

- Listing of read literature and main contribution / idea
- Build the foundation of the related work part of the project and thesis